Telecommunication Systems I

Exercise 3

Submission Date: May 22, 2026

The assignment may be submitted by teams of $\leq$ two students

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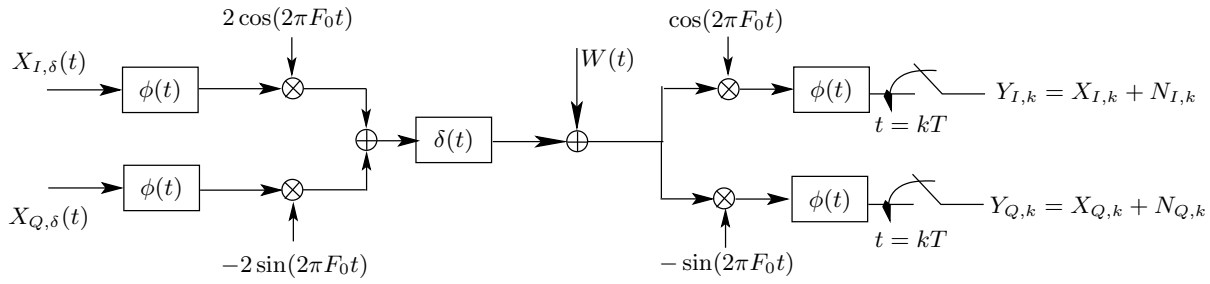
Points 130/300



In this exercise, we will simulate the telecommunication system shown in the Figure, assuming that 16-QAM modulation is used, and we will study its performance. More specifically, we will implement the 16-QAM system as two independent 4-PAM systems, with constellation $\{\pm A, \pm 3A\}$.

1. For a given N (for example, $N = 200$), generate a binary sequence with $4N$ equiprobable bits.

2. (15) Write a function

   $$\text{function} \quad \text{X} = \text{bits_to_4_PAM}(\text{bit_seq}, \text{A})$$

   which, using Gray coding, maps the input binary sequence bit_seq to a sequence of 4-PAM symbols.

3. Map the first $2N$ bits of the sequence from Step 1 to the 4-PAM symbols $X_{I,n}$, for $n = 1, \ldots, N$, and the next $2N$ bits to the 4-PAM symbols $X_{Q,n}$, for $n = 1, \ldots, N$.

4. (5) Pass the sequences $\{X_{I,n}\}$ and $\{X_{Q,n}\}$ through the SRRC pulse-shaping filters. Assume a symbol period $T = 10^{-3}$ sec, over $= 10$, $T_s = \frac{T}{\text{over}}$ and construct and plot the output waveforms $X_I(t)$ and $X_Q(t)$ (**set the correct time axis**), as well as their periodograms.

5. (5) Multiply the waveforms $X_I(t)$ and $X_Q(t)$ by the corresponding carriers (for example, $F_0 = 2\,\text{KHz}$) and generate the waveforms

$$X_I^{\text{mod}}(t) = 2\,X_I(t)\cos(2\pi F_0 t),\;\; X_Q^{\text{mod}}(t) = -2\,X_Q(t)\sin(2\pi F_0 t).$$

Plot the resulting waveforms as well as the corresponding periodograms. What do you observe?

6. (5) Construct and plot the channel input,

$$X^{\text{mod}}(t) = X_I^{\text{mod}}(t) + X_Q^{\text{mod}}(t)$$

and its periodogram. What do you observe?

7. Assume that the channel is ideal.

8. (5) At the channel output, add white Gaussian noise $W(t)$ with variance equal to

$$\sigma_W^2 = \frac{10A^2}{T_s \cdot 10^{\frac{\text{SNR}_{\text{dB}}}{10}}}.$$

Note: it can be shown that, in this case, the random variables $N_{I,n}, N_{Q,n}$, $n = 0, \ldots, N-1$, are independent with

$$N_{I,n}, N_{Q,n} \sim \mathcal{N}(0, \sigma_N^2), \quad \text{where } \sigma_N^2 = \frac{T_s\,\sigma_W^2}{2}.$$

In this way, you ensure that the SNR *at the output of the matched filter, measured in* dB*, is* SNR_{dB} *(for example,* $\text{SNR}_{\text{dB}} = 10, 20$*), because*

$$10\log_{10}\frac{P_X}{P_N} = 10\log_{10}\frac{10A^2}{2\sigma_N^2} = \text{SNR}_{\text{dB}}.$$

9. (5) At the receiver, split the noisy waveform into two branches and multiply them by the carriers $\cos(2\pi F_0 t)$ and $-\sin(2\pi F_0 t)$, respectively. Plot the waveforms that result and their periodograms. What do you observe?

10. (5) Pass the waveforms computed in the previous step through the matched filters. Plot the resulting waveforms and their periodograms (**set the correct time axis**). What do you observe?

11. (5) Sample the output of the matched filters at the appropriate time instants and plot the output sequence using the `scatterplot` command.

12. (10) Write a function

$$\text{function} \quad \text{est_X} = \text{detect_4_PAM}(\text{Y}, \text{A})$$

which uses the nearest-neighbor rule and makes decisions for the input sequence of 4-PAM symbols on a symbol-by-symbol basis. Apply the function separately to the samples obtained at the inphase output and at the quadrature output, and obtain decisions for the corresponding input sequences.

13. (10) Using the input sequences and the decisions, compute the **number of symbol decision errors for the** 16**-QAM constellation.**

14. (10) Write a function

$$\text{function} \quad \text{est_bit} = \text{PAM_4_to_bits}(\text{X}, \text{A})$$

which uses the inverse Gray mapping, that is, the mapping from symbols to pairs of bits, and, from the decisions for the input symbol sequences, computes the estimated input binary sequence.

15. (10) Compute the number of bit errors.

If you have implemented the steps above correclty, you have understood how a system that uses 16-QAM modulation operates (of course, we have assumed that the channel is ideal and that we are perfectly synchronized).

In the second part, you will estimate the symbol and bit error probabilities using the Monte Carlo method.

1. For $\text{SNR}_{\text{dB}} = [0 : 2 : 16]$, experimentally compute the symbol-decision and bit error probabilities by repeating the steps above K times (for example, $K = 200, 1000$) for each SNR. The experimental estimate of the symbol error probability for each SNR is computed as follows:

$$\hat{P}(\mathtt{E}_{\text{symbol}}) = \frac{\text{total number of symbol decision errors}}{\text{total number of transmitted symbols}},$$

whereas the bit error probability is computed as follows:

$$\hat{P}(\mathtt{E}_{\text{bit}}) = \frac{\text{total number of bit decision errors}}{\text{total number of transmitted bits}}.$$

2. (20) Plot, using semilogy, the theoretical and experimental symbol error probabilities as a function of SNR_{dB}. What do you observe?

3. (20) Plot, using semilogy, the experimental bit error probability as a function of SNR_{dB}. In the same figure, also plot the theoretical approximation for the bit error probability. What do you observe? Can you explain the phenomenon?